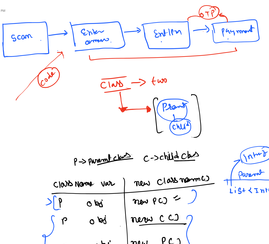

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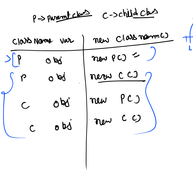
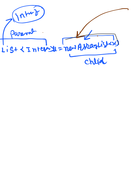
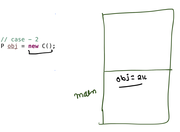
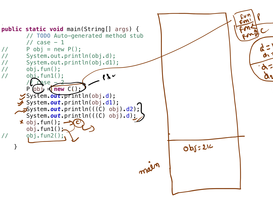
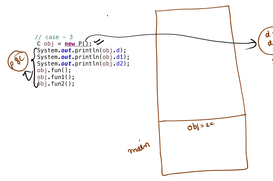
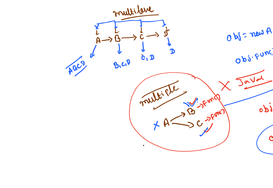




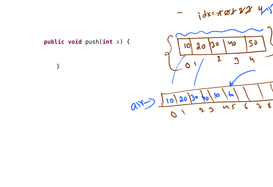









```
public class Stack {
    private int[] arr;
    private int idx = -1;

    public Stack() {
        // TODO Auto-generated constructor stub
        this(5);
    }

    public Stack(int n) {
        // TODO Auto-generated constructor stub
        arr = new int[n];
    }

    // O(1)
    public boolean isEmpty() {
        return idx == -1;
    }

    // O(1)
    public void push(int item) throws Exception {
        if (isFull()) {
            throw new Exception("Bklol Vidyarthi Stack full hogya hai");
        }
        arr[++idx] = item;
    }

    // O(1)
    public int pop() throws Exception {
        if (isEmpty()) {
            throw new Exception("Bklol Vidyarthi Stack Empty  hai");
        }
        int x = arr[idx--];
        return x;
    }

    // O(1)
    public int peek() throws Exception {
        if (isEmpty()) {
            throw new Exception("Bklol Vidyarthi Stack Empty  hai");
        }
        int x = arr[idx];
        return x;
    }

    // O(1)
    public int size() {
        return idx + 1;
    }

    // O(1)
    public boolean isFull() {
        return size() == arr.length;
    }

    public void Display() {
        for (int i = 0; i <= idx; i++) {
            System.out.print(arr[i] + " ");
        }
    }
}
```




```
public class Queue {
    private int[] arr;
    private int front = 0;
    private int size = 0;// size rear jaisa and number of element bhi batayega

    public Queue() {
        // TODO Auto-generated constructor stub
        this(5);
    }

    public Queue(int n) {
        // TODO Auto-generated constructor stub
        arr = new int[n];
    }

    // O(1)
    public boolean isEmpty() {
        return size == 0;
    }

    // O(1)
    public int size() {
        return size;
    }

    // O(1)
    public boolean isFull() {
        return size == arr.length;
    }

    // O(1)
    public void Enqueue(int item) throws Exception {
        if (isFull()) {
            throw new Exception("Bklol Vidyarthi Queue full hogya hai");
        }
        int idx = (front + size) % arr.length;
        arr[idx] = item;
        size++;
    }

    // O(1)
    public int Dequeue() throws Exception {
        if (isFull()) {
            throw new Exception("Bklol Vidyarthi Queue Empty hogya hai");
        }
        int item = arr[front];
        front = (front + 1) % arr.length;
        size--;
        return item;
    }

    // O(1)
    public int getFront() throws Exception {
        if (isFull()) {
            throw new Exception("Bklol Vidyarthi Queue Empty hogya hai");
        }

        int item = arr[front];

        return item;
    }

    // O(1)
    public void Display() {
        for (int i = 0; i < size; i++) {
            int idx = (front + i)%arr.length;
            System.out.print(arr[idx]+" ");
        }
    }
}
```